

Math 421/510: Homework 6

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1. Let $1 \leq p \leq 2$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then $\mathcal{F} : L^p(\mathbb{R}^n) \rightarrow L^q(\mathbb{R}^n)$ and

$$\|\hat{f}\|_q \leq \|f\|_p.$$

Solution.

By Plancherel's theorem, we have that for $f \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, that $\|\mathcal{F}(f)\|_2 = \|f\|_2$. Furthermore, by theorem 5.10.1 in the notes, $\|\mathcal{F}(f)\|_\infty \leq \|f\|_1$. Since $\mathcal{F}$ is a linear operator we apply Theorem 5.51 in the notes (Riesz Thorin Interpolation theorem) with $p_0 = 1$, $p_1 = 2$, $q_0 = \infty$, $q_1 = 2$, $M_0 = M_1 = 1$, so that for

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1},$$

we have that $\mathcal{F}$ extends to a bounded linear operator $\mathcal{F} : L^{p_\theta}(\mathbb{R}^n) \rightarrow L^{q_\theta}(\mathbb{R}^n)$ with $\|\mathcal{F}(f)\|_{L^{q_\theta}} \leq M_0^{1-\theta} M_1^\theta \|f\|_{L^{p_\theta}} = \|f\|_{L^{p_\theta}}$, for $0 \leq \theta \leq 1$.

Now, let $1 \leq p \leq 2$. Then, in particular, we have that when $\theta = 2(1-p)$, then

$$p_\theta = \left(\frac{1 - 2(1 - \frac{1}{p})}{1} + \frac{2(1 - \frac{1}{p})}{2} \right)^{-1} = \left(\frac{1}{p} \right)^{-1} = p.$$

Also, then

$$\frac{1}{q_\theta} = \frac{\theta}{q_1} = \frac{2(1 - \frac{1}{p})}{2} = 1 - \frac{1}{p},$$

so that $\frac{1}{p} + \frac{1}{q} = 1$. Therefore, for $1 \leq p \leq 2$, we have that $\mathcal{F} : L^p(\mathbb{R}^n) \rightarrow L^q(\mathbb{R}^n)$ and $\|\hat{f}\|_q \leq \|f\|_p$, as desired.

2. For each integer $m \geq 2$ there exists a constant $C_m > 0$ such that the following holds.

If $f \in C^m(a, b)$ and

$$f^{(m)}(x) \geq 1 \quad \text{for all } x \in (a, b),$$

then for every $R \geq 1$,

$$\left| \int_a^b e^{iRf(x)} dx \right| \leq C_m R^{-1/m}.$$

More generally, if $f^{(m)}(x) \geq \lambda > 0$ on (a, b) , then

$$\left| \int_a^b e^{iRf(x)} dx \right| \leq C_m (R\lambda)^{-1/m}.$$

Moreover, the decay rate $R^{-1/m}$ is sharp in general: one cannot replace $R^{-1/m}$ by $R^{-\alpha}$ for any $\alpha > \frac{1}{m}$.

Solution.

We prove by induction.

We start with the base case, for $m = 2$, which we proved in class. For the generalization, if $f^{(2)} \leq \lambda > 0$ on (a, b) , then $\frac{1}{\lambda} f^{(m)}(x) \geq 1$ for all (a, b) , so the second Van der Corput lemma quickly gives us that

$$\left| \int_a^b e^{iRf(x)} dx \right| = \left| \int_a^b e^{i(\lambda R)(\frac{1}{\lambda} f(x))} dx \right| \leq \frac{10}{\sqrt{\lambda R}},$$

as required. This completes our base case.

For inductive step, suppose that the result holds for $m - 1$, where $m \geq 3$. That is, if $f^{(m-1)}(x) \geq 1$ for all $x \in (a, b)$, then there exists some C_{m-1} such that

$$\left| \int_a^b e^{iRf(x)} dx \right| \leq C_{m-1} R^{-1/(m-1)}.$$

Now, suppose $f^{(m)}(x) \geq 1$ for all $x \in (a, b)$. In particular, this implies that $f^{(m-1)}$ is strictly increasing. If $f^{(m-1)} \geq 1$, then by our inductive hypothesis, we have

$$\left| \int_a^b e^{iRf(x)} dx \right| \leq C_{m-1} R^{-1/(m-1)} \leq C_{m-1} R^{-1/m},$$

as required. Next, suppose that $f^{(m-1)}(x) \geq \lambda > 0$ in (a, b) , for any $\lambda > 0$. Since $f^{(m-1)}$ is strictly increasing, we must have that $f^{(m-1)}(x) \geq \lim_{y \rightarrow a} f^{(m-1)}(y)$ for $x \in (a, b)$. In particular, the mean value theorem gives us that $f^{(m-1)}(a+\delta) > \lambda + \delta$, for $0 < \delta < b-a$. Choose $\delta > \frac{1}{R^{1/m}}$. Then, where the bound on the second integral follows by the inductive hypothesis, and the bound on the first integral follows from the fact that $|e^{iRf(x)}| = 1$,

$$\begin{aligned} \left| \int_a^b e^{iRf(x)} dx \right| &\leq \left| \int_a^{a+\delta} e^{iRf(x)} dx \right| + \left| \int_{a+\delta}^b e^{iRf(x)} dx \right| \leq \frac{1}{R^{1/m}} + C_{m-1} (R \frac{1}{R^{1/m}})^{-1/(m-1)} \\ &= \frac{1}{R^{1/m}} + C_{m-1} (R^{\frac{m-1}{m}})^{-1/(m-1)} \\ &= (1 + C_{m-1}) R^{-1/m}. \end{aligned}$$

Finally, suppose that there exists some $x_0 \in (a, b)$ such that $f^{(m-1)}(x_0) = 0$. Since $f^{(m-1)}$ is strictly increasing, there can only be one such point. Also, by the mean value theorem, if $\delta > 0$, then we have that both

$$\begin{aligned} f^{(m-1)}(x_0 - \delta) &\leq 0 - \delta \text{ and} \\ f^{(m-1)}(x_0 + \delta) &\geq 0 + \delta. \end{aligned}$$

Then, we can split up the integral into three intervals:

$$\int_a^{x_0-\delta} e^{iRf(x)} dx + \int_{x_0-\delta}^{x_0+\delta} e^{iRf(x)} dx + \int_{x_0+\delta}^b e^{iRf(x)} dx = I + II + III.$$

We compute bounds on each in turn. For I , note that if we take the conjugate of the integral, its magnitude doesn't change, and we get

$$\bar{I} = \overline{\int_a^{x_0-\delta} e^{iRf(x)} dx} = \int_a^{x_0-\delta} e^{-iRf(x)} dx.$$

Then, since $(-f)^{(m-1)} = -(f^{(m-1)}) \geq -(-\delta) = \delta$ for $x \in (a, x_0 - \delta)$ so we can apply our inductive hypothesis to get

$$|I| = |\bar{I}| \leq C_{m-1}(R\delta)^{-1/(m-1)}.$$

For II we use $|e^{iRf(x)}| = 1$ to get that

$$\left| \int_{x_0-\delta}^{x_0+\delta} e^{iRf(x)} dx \right| \leq 2\delta.$$

Finally, for III , we use our inductive hypothesis, along with the bound $f^{(m-1)}(x_0+\delta) \geq \delta$ that we established earlier:

$$|III| \leq C_{m-1}(R\delta)^{-1/(m-1)}.$$

Thus,

$$\begin{aligned} \int_a^{x_0-\delta} e^{iRf(x)} dx + \int_{x_0-\delta}^{x_0+\delta} e^{iRf(x)} dx + \int_{x_0+\delta}^b e^{iRf(x)} dx &= I + II + III \\ &\leq 2C_{m-1}(R\delta)^{-1/(m-1)} + 2\delta. \end{aligned}$$

Now, if we set $\delta = \frac{1}{R^{1/m}}$, we get:

$$\left| \int_a^b e^{iRf(x)} dx \right| \leq 2C_{m-1}(R^{\frac{m-1}{m}})^{-1/(m-1)} + \frac{2}{R^{1/m}} = (2C_{m-1} + 2)R^{-1/m},$$

as desired. (The desired constant is $\max(2C_{m-1} + 2)$). For the general case, consider that if $f^{(m)} \geq \lambda > 0$, then $1/\lambda f^{(m)} \geq 1$, so we can use the result we just derived to compute

$$\left| \int_a^b e^{iRf(x)} dx \right| = \left| \int_a^b e^{i(R\lambda)(f(x)/\lambda)} dx \right| \leq C_m(\lambda R)^{-1/m},$$

as desired.

To show that the decay rate is strict, consider the function $f(x) = \frac{1}{m!}x^m$. In particular, we have $f^{(m)}(x) = 1$. Then, consider:

$$\int_0^1 e^{iR\frac{1}{m!}x^m} dx = R^{-1/m} \int_0^{R^{1/m}} e^{i\frac{1}{m!}u^m} du.$$

Now, as $R \rightarrow \infty$, we can use a similar contour integration trick as shown in the extra lecture notes (but since this is not a complex analysis course, I will omit the details) to show that

$$\int_0^{R^{1/m}} e^{i\frac{1}{m!}u^m} du = \int_0^{R^{1/m}} e^{-\frac{1}{m!}u^m} du + \varepsilon(R),$$

where $\varepsilon(R) \rightarrow 0$ as $R \rightarrow \infty$.

Now make the change of variables

$$t = \frac{1}{m!} u^m.$$

Then

$$dt = \frac{m}{m!} u^{m-1} du = \frac{1}{(m-1)!} u^{m-1} du.$$

Since $u = (m!t)^{1/m}$, we have

$$du = (m-1)! u^{1-m} dt = (m-1)! (m!t)^{\frac{1}{m}-1} dt = \frac{(m!)^{1/m}}{m} t^{\frac{1}{m}-1} dt.$$

Also, when $u = 0$ we have $t = 0$, and when $u = R^{1/m}$ we have $t = R/m!$. Therefore,

$$\int_0^{R^{1/m}} e^{-\frac{1}{m!} u^m} du = \frac{(m!)^{1/m}}{m} \int_0^{R/m!} t^{\frac{1}{m}-1} e^{-t} dt.$$

Then we cite Rudin, in the section on the Gamma function, which says that

$$\Gamma\left(\frac{1}{m}\right) = \int_0^\infty t^{\frac{1}{m}-1} e^{-t} dt.$$

Hence

$$\lim_{R \rightarrow \infty} \int_0^{R^{1/m}} e^{-\frac{1}{m!} u^m} du = \frac{(m!)^{1/m}}{m} \Gamma\left(\frac{1}{m}\right),$$

which is a finite real number. It follows that

$$\int_0^{R^{1/m}} e^{i\frac{1}{m!} u^m} du \rightarrow \frac{(m!)^{1/m}}{m} \Gamma\left(\frac{1}{m}\right) \quad \text{as } R \rightarrow \infty.$$

Therefore, we have that

$$\int_0^1 e^{iR\frac{1}{m!} x^m} dx = R^{-1/m} \int_0^{R^{1/m}} e^{i\frac{1}{m!} u^m} du \rightarrow R^{-1/m} \left(\frac{(m!)^{1/m}}{m} \Gamma\left(\frac{1}{m}\right) \right),$$

so the decay rate is strict.

3. Let $f_n := \chi_{[2^n, 2^{n+1}]}$ for $n \in \mathbb{N}$.

- (a) Let $1 < p < \infty$. Is it true that $f_n \rightarrow 0$ weakly in $L^p(\mathbb{R})$ as $n \rightarrow \infty$? Explain why or why not.
- (b) Is it true that $f_n \rightarrow 0$ weak* in $L^\infty(\mathbb{R}) = (L^1(\mathbb{R}))^*$ (i.e. as linear functionals on L^1) as $n \rightarrow \infty$? Explain why or why not.

Solution.

(a) This is false. To show that is this is the case, consider the function

$$g = \sum_{n=1}^{\infty} 2^{-n} \chi_{[2^n, 2^{n+1})}.$$

To see that $\varphi_g \in (L^p)^* \cong L^q$, we consider that

$$\int_{\mathbb{R}} |g|^q = \sum_{n=1}^{\infty} 2^{-nq} 2^n = \sum_{n=1}^{\infty} (2^{1-q})^n.$$

Since $q = (1 - \frac{1}{p})^{-1}$, we have that $1 < q < \infty$, so that $1 - q < 0$, and $0 < 2^{1-q} < 1$, so that this sum is a geometric series and so converges.

Then, computing the linear functional for each f_n , we have

$$\varphi_g(f_n) = \int_{\mathbb{R}} g f_n = \int_{2^n}^{2^{n+1}} 2^{-n} = 1,$$

which does not go to zero as $n \rightarrow \infty$. Therefore, it is not true that $f_n \rightarrow 0$ weakly in $L^p(\mathbb{R})$ as $n \rightarrow \infty$.

(b) Fix $g \in L^1$. Then, as a linear functional, we have that $f_n(g) = \int_{\mathbb{R}} f_n g$. By considering that the finitely supported simple functions are dense in L^1 , we find some $h \in L^1$ such that h is a finitely supported simple function arbitrarily close the g in the $L^1(\mathbb{R})$ norm and n is such that $n > N$ for $2^N > \sup\{\text{supp}(h)\}$. It's quick to see that $\|f_n\|_{L^\infty(\mathbb{R})} = 1$. Then,

$$\begin{aligned} \left| \int_{\mathbb{R}} f_n g \right| &\leq \left| \int_{\mathbb{R}} f_n (g - h) \right| + \left| \int_{\mathbb{R}} f_n h \right| \leq \|f_n\|_{L^\infty(\mathbb{R})} \|g - h\|_{L^1(\mathbb{R})} + 0 \\ &< 1 \cdot \varepsilon + 0 = \varepsilon. \end{aligned}$$

Since ε was arbitrary we conclude $f_n \rightarrow 0$ weak*.